

## 3.1: Causal Inference

What is it?

using results from **stats analysis** to **make inference** about the **causal effect** of an **exposure/treatment** on an outcome.

Notation:

$Y$  = outcome

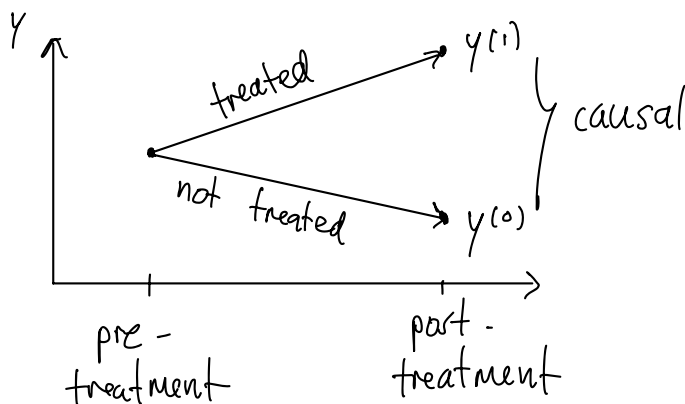
$Z$  = treatment / exposure. // binary. 1 = exposed,

$X$  = baseline or pretreatment variable.

Potential outcome notation:

$y^{(1)}$  = outcome when  $Z = 1$  // under treatment.

$y^{(0)}$  = outcome when  $Z = 0$ , // under control.



for an individual,  
 $y^{(1)} - y^{(0)} = \text{causal effect.}$

**Average treatment effect in the population (ATE)**

answers the question:

How effective would the treatment have been in the population?

$$\text{ATE} = E(y^{(1)} - y^{(0)})$$

Average Treatment Effect in the Treated (ATT):

How would those who received treatment have done had they received the comparison condition?

$$ATT = E(y^{(1)} - y^{(0)} | Z = 1)$$

// more of interest if treatment is offered to everyone.

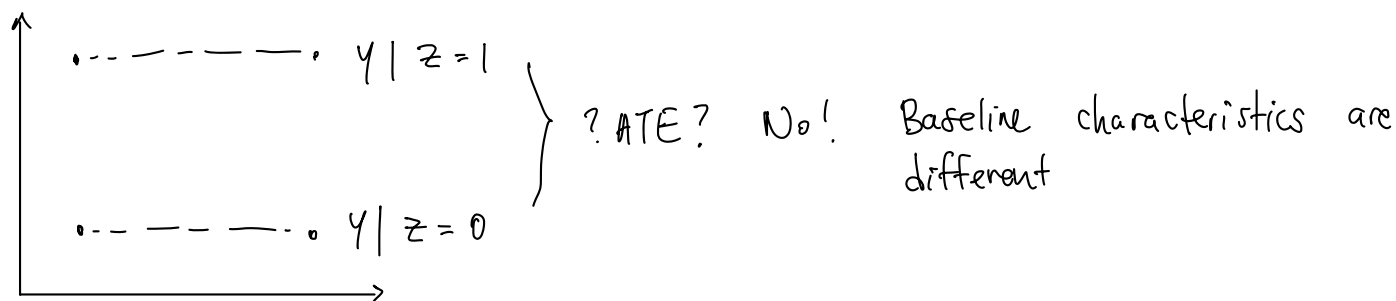
Randomized Studies: gold standard,

Randomization **balances** the baseline characteristics,

→ strong ignorability → treatment assignment is independent of potential outcomes.

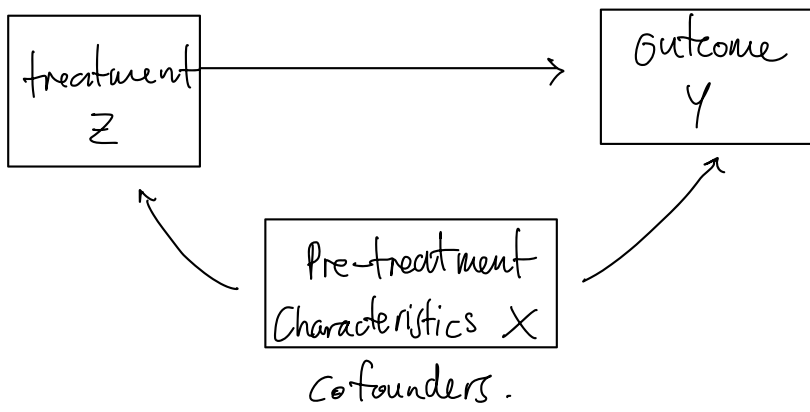
Observational Studies:

- provides a way to get at causal effects.
- However, groups are usually **imbalanced**.
- selection bias = biggest challenge.



They may be different **before** the treatment rather than from the treatment!

Selection bias occurs when these differences are **associated** with the outcome.



DAG : directed acyclic graph.

## 3.2: Propensity Scores

How to handle selection bias?

### Option 1: Regression Adjustment:

- Adjust a regression examining the effect of  $Z$  on  $Y$  for  $X$ .
- Valid as long as the regression model is correctly specified and  $X$  includes all confounders.

### Option 2: Propensity Score:

- probability of an individual receiving treatment given pre-treatment characteristics / observed covariates.

$$p(x) = P(Z = 1 | X)$$

↑ pre-treatment characteristics.

- used to create balance btwn the treatment and control group. Mimics randomization.
- It can be used for estimating causal effects as long as we assume the 2 following:

#### Assumptions

- ①  $y^{(1)}, y^{(0)} \perp Z | X \Rightarrow$  there are no unmeasured confounders.
- ②  $0 < p(x) < 1 \Rightarrow$  there is overlap btwn groups.

- Once you obtain propensity score, how to use it?
  - stratification
  - matching
  - weighing (focus of this class).

ATE can be estimated using propensity score weighting:

$$ATE = E(Y^{(1)} - Y^{(0)})$$

$$Y^{(1)} = \text{treatment}$$

$$Y^{(0)} = \text{control}$$

// this is IPW of ATE.

$$= \frac{\sum_{i: z_i=1} Y_i W_i}{\sum_{i: z_i=1} W_i} - \frac{\sum_{i: z_i=0} Y_i W_i}{\sum_{i: z_i=0} W_i}, \quad \text{where}$$

$X_i$  = covariates for individual  $i$

$Y_i$  = outcome for individual  $i$

$z_i$  = treatment indicator (1 if treated, else 0).

$$W_i = \begin{cases} 1/p(X_i) & \text{if } z_i = 1 \\ 1/(1-p(X_i)) & \text{if } z_i = 0 \end{cases} \quad \text{// weighs each person depending on likelihood of receiving treatment,}$$

$p(X_i)$  = propensity score

$$f(x) \quad p(x | z=1)$$

$$w \cdot p(x | z=1) = p(x)$$

$$\Rightarrow \frac{w \cdot p(z=1 | x) \cdot p(x)}{p(z=1)} = p(x) \quad \text{// Bayes}$$

$$\Rightarrow w = \frac{p(z=1)}{p(z=1 | x)} = \frac{1}{p(z=1 | x)} = \frac{1}{p(x)}$$

$$\widetilde{ATE} = \frac{1}{n_1} \sum_{i: z_i=1} y_i - \frac{1}{n_0} \sum_{j: z_j=1} y_j \quad // \text{ wrong approach.}$$

$n_1$  = # of people in the treatment group

$n_0$  = # people in control group.

## How to estimate propensity score?

$$Z \sim X$$

logistic regression

$$P(Z = 1 \mid X)$$

## machine learning methods for estimating propensity scores:

- achieve better balance btwn the 2 groups
- reduce bias in treatment effect estimates
- produce more stable propensity scores, better than logistic regression.

## Many methods to estimate propensity scores:

- Covariate balancing
- super learning
- high dimensional propensity score
- entropy balancing.

We will focus on generalized boosted models (GBM) to estimate propensity score weights.

- nonparametric approach.
- outcome can be binary, discrete, continuous.
- allows for interactions among covariates
- invariant to monotonic transformation of covariates.

GBM models the log odds of treatment assignment:

$$g(x) = \log \left\{ \frac{p(x)}{1-p(x)} \right\} \quad \text{by initializing}$$

$$g(x) = \log \left\{ \frac{\tilde{z}}{1-\tilde{z}} \right\}, \quad \text{where } \tilde{z} \text{ // constant baseline log odds of assignment to treatment.}$$

$\tilde{z}$  = avg treatment assignment indicator for the entire sample.

→ searches for a small adjustment,  $h(x)$ , to add to initial estimate.

→ Fit is measured w/ Bernoulli log likelihood:

$$l(g) = \sum_i z_i g(x_i) - \log \{ 1 + \exp(g(x_i)) \}, \quad \text{where}$$

$$z_i = \begin{cases} 0, & \text{when } g(x_i) < 0 \\ 1, & \text{when } g(x_i) \geq 0 \end{cases} \quad \text{// larger } l(g) = \text{better fit.}$$

→ If adjustment improves fit, then current model becomes  $g(x) + h(x)$ .

$h(x)$  = regression tree that models the residuals from the current fit

GBM = sum of simple regression trees

"number of trees" = iterations.

→ balance → optimal # of trees

| Treatment | Control |
|-----------|---------|
| 22        | 65      |
| 50%       | 15%     |
| 50%       | 52%     |



## Metrics to Assess Balance:

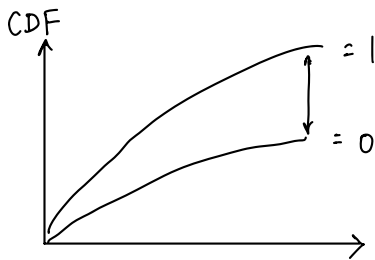
- p-value from a test (t-test).
- standardized effect size (ES) difference btwn the 2 groups.  
= difference in the means divided by the standard deviation,  
⇒ threshold 0.2 is used.

We want the difference to be small.

$$ES < 0.2$$

< 0.1 even better.

- Kolmogorov-Smirnov (KS) statistics:
  - The maximal vertical distance btwn the empirical cumulative distribution functions of two samples.



twang = R toolkit for weighting and analysis of nonequivalent groups

gbm = generalized boosted models.

## Steps to estimate the treatment effects.

- ① identify potential confounders.
- ② Assess balance w/o propensity score weighting
- ③ Estimate propensity scores (GBM, ML methods, etc).
- ④ Assess balance of weighted groups (KS statistics, t-test, etc).
- ⑤ If balance is achieved, estimate the treatment effect using the propensity score weighting (matching, stratification, IPW, etc).
- ⑥ Interpret within context of required assumptions.

### 3.3: Double Robust Estimation

Both regression adjustment and propensity score weighting relies on assumption that there are no unmeasured confounders:  $y^{(1)}, y^{(0)} \perp Z | X$ .

Regression adjustment  $\rightarrow$  valid treatment effect estimate if your outcome model is correct. If incorrect  $\rightarrow$  biased estimate.

Example: Suppose you postulate this model:

$$E(Y) = B_0 + B_1 Z + B_2 X$$

You think the model is  $\underbrace{Y}_{\text{outcome}} \sim \underbrace{Z + X}_{\text{predictors}}$ ,

It can be shown that  $\beta_1$  is an estimate of ATE:

$$\Rightarrow \text{ATE} = E(Y^{(1)} - Y^{(0)})$$

$$= E(Y^{(1)}) - E(Y^{(0)})$$

unmeasured  
confounders

$$\begin{aligned} &= E_X[E(Y^{(1)} | X)] - E_X[E(Y^{(0)} | X)] \\ &= E_X[E(Y | X, Z=1)] - E_X[E(Y | X, Z=0)] \end{aligned}$$

model  
holds

$$= E_X[B_0 + B_1 + B_2 X] - E_X[B_0 + B_2 X]$$

$$= \cancel{B_0} + B_1 + B_2 \cancel{E_X(X)} - \cancel{B_0} - B_2 \cancel{E_X(X)}$$

$$= \beta_1$$

However, let's say the true model is

$$E(Y) = \alpha_0 + \alpha_1 Z + \alpha_2 X^2.$$

By similar algebra you can show that

$$ATE = \alpha_1. \text{ In general } \beta_1 \neq \alpha_1.$$

Therefore your estimate of ATE will be incorrect.

Propensity Score weighting:

If the model used to estimate the propensity scores is correctly specified, then propensity score weighing will result in a valid treatment effect estimate.

If model is not correct, you may get a biased estimate.

Recall the propensity score weighted estimate of ATE:

$$\hat{ATE} = \frac{\sum_{i: z_i=1} Y_i W_i}{\sum_{i: z_i=1} W_i} - \frac{\sum_{i: z_i=0} Y_i W_i}{\sum_{i: z_i=0} W_i}, \quad \text{where}$$

$$W_i = \begin{cases} 1/p(x_i) & \text{if } z_i = 1 \\ 1/(1-p(x_i)) & \text{if } z_i = 0 \end{cases}.$$

This is called the Inverse Probability Weighted (IPW) estimate of the average treatment effect (ATE).

If the model used to obtain the propensity scores:  $p(X) = p(Z=1|X)$  is correct, then under the assumption of no measured confounders, it can be shown that:

$$E(\widehat{ATE}) = E(Y^{(1)} - Y^{(0)}) = ATE$$

However, if the model is not correct then  $E(\widehat{ATE}) \neq ATE$  and the estimate will not be correct.

Double Robust Estimation is an estimation method that combines these 2 ideas, resulting in a valid estimate as long as at least one of {propensity score model, outcome model} is correctly specified.

Hajek estimator: a double robust estimator.

$$\begin{aligned}\widehat{ATE}_{DR} &= \frac{1}{n} \left[ \sum_{i: Z_i=1} Y_i w_i - \sum_{j: Z_j=0} Y_j w_j \right] \\ &= \frac{1}{n} \sum_{i=1}^n Z_i Y_i w_i - \frac{1}{n} \sum_{i=1}^n (1 - Z_i) Y_i w_i \\ &= \frac{1}{n} \sum_{i=1}^n \frac{Z_i Y_i}{p(X_i)} - \frac{1}{n} \sum_{i=1}^n \frac{(1 - Z_i) Y_i}{1 - p(X_i)} \\ &\quad \underbrace{\qquad\qquad\qquad}_{E(\uparrow) = E(Y^{(1)})} \quad \underbrace{\qquad\qquad\qquad}_{E(\uparrow) = E(Y^{(0)})}\end{aligned}$$

$$E\left(\frac{A}{B}\right) \neq \frac{E(A)}{E(B)}$$

Proof:

$$E\left[\frac{1}{n} \sum_{i=1}^n \frac{Z_i Y_i}{p(X_i)}\right] = E\left[\frac{ZY}{p(X)}\right]$$

Note:

$$\begin{aligned}ZY &= Z \{ ZY^{(1)} + (1-Z)Y^{(0)} \} \\&= Z^2 Y^{(1)} + Z(1-Z)Y^{(0)} \\&= ZY^{(1)}\end{aligned}$$

$$\Rightarrow E\left(\frac{ZY}{p(x)}\right) = E\left(\frac{ZY^{(1)}}{p(x)}\right) = E_x\left[E\left(\frac{ZY^{(1)}}{p(x)} \mid X, Y^{(1)}\right)\right]$$

$$= E\left[\frac{Y^{(1)}}{p(x)} E(Z|X)\right]$$

$$= E\left[\frac{Y^{(1)}}{p(x)} p(Z=1|X)\right]$$

$$= E\left[\frac{Y^{(1)}}{\cancel{p(x)}} \cdot \cancel{p(x)}\right]$$

$$= E[Y^{(1)}]$$

outcome model:

holds

does not hold

holds

does not hold

propensity score model:

holds

holds

does not hold

does not hold

double robust estimator:

OK

OK

OK.

Not OK.

$$\begin{aligned}\hat{ATE}_{DR} &= \frac{1}{n} \sum_{i=1}^n \left[ \frac{z_i y_i}{p(x_i)} - \frac{\{z_i - p(x_i)\}}{p(x_i)} m(x_i, z_i) \right] \\&\quad - \frac{1}{n} \sum_{i=1}^n \left[ \frac{(1-z_i) y_i}{1-p(x_i)} - \frac{\{z_i - p(x_i)\}}{1-p(x_i)} m(x_i, z_i) \right]\end{aligned}$$

$$= \hat{\mu}_{1,DR} - \hat{\mu}_{0,DR}, \text{ where}$$

$p(x_i)$  = estimate from propensity score model,

$m(x_i, z_i)$  = estimate of  $y$  from the outcome model.

$$m(x_i, z_i) = \beta_0 + \beta_1 z + \beta_2 x$$

$$m_1(x_i, \hat{\beta}) = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 x_i$$

$$m_0(x_i, \hat{\beta}) = \hat{\beta}_0 + \hat{\beta}_2 x_i$$

$$E(\hat{ATE}_{DR}) = E(y^{(1)} - y^{(0)}) = ATE$$

$$E(\hat{\mu}_{1, DR}) = E(y^{(1)})$$

Proof:

$$E(\hat{\mu}_{1, DR}) = E\left[\frac{1}{n} \sum_{i=1}^n \left[ \frac{z_i y_i}{p(x_i)} - \frac{[z_i - p(x_i)]}{p(x_i)} m_1(x, \hat{\beta}) \right]\right]$$

$$= E\left[\frac{zy}{p(x)} - \frac{z - p(x)}{p(x)} m_1(x, \hat{\beta})\right]$$

$$= E\left[y^{(1)} - y^{(1)} + \frac{zy^{(1)}}{p(x)} - \frac{z - p(x)}{p(x)} m_1(x, \hat{\beta})\right]$$

$$= E\left(y^{(1)} - \frac{y^{(1)} p(x)}{p(x)} + \frac{zy^{(1)}}{p(x)} - \frac{z - p(x)}{p(x)} m_1(x, \hat{\beta})\right)$$

$$= E\left(y^{(1)} + \frac{y^{(1)} [z - p(x)]}{p(x)} - \frac{z - p(x)}{p(x)} m_1(x, \hat{\beta})\right)$$

$$= E\left(y^{(1)} + \frac{z - p(x)}{p(x)} [y^{(1)} - m_1(x, \hat{\beta})]\right)$$

$$= E(y^{(1)}) + E\left(\frac{z - p(x)}{p(x)} [y^{(1)} - m_1(x, \hat{\beta})]\right)$$

?

$$E(z=1|x) = p(x) \quad E(y|z, x) = m(x, z)$$

$$\begin{aligned} & E\left(\frac{z - p(x)}{p(x)} [y^{(1)} - m_1(x, \hat{\beta})]\right) \\ &= E\left(E\left(\frac{z - p(x)}{p(x)} (y^{(1)} - m_1(x, \hat{\beta})) \mid y^{(1)}, x\right)\right) \\ &= E\left([y^{(1)} - m_1(x, \hat{\beta})] \frac{E(z|y^{(1)}, x) - p(x)}{p(x)}\right) \\ &= E\left(y^{(1)} - m_1(x, \hat{\beta}) \underbrace{\frac{p(x) - p(x)}{p(x)}}_0\right) \end{aligned}$$

Let's say outcome model is correct, but propensity model is wrong:

$$E(z=1|x) = p(x) \quad E(y|z, x) = m(x, z)$$

$$\begin{aligned} & E\left[\frac{z - p(x)}{p(x)} (y^{(1)} - m_1(x, \hat{\beta}))\right] \\ &= E\left(E\left(\frac{z - p(x)}{p(x)} [y^{(1)} - m_1(x, \hat{\beta})] \mid z, x\right)\right) \\ &= E\left(\frac{z - p(x)}{p(x)} \underbrace{\left[E(y^{(1)} \mid z, x) - m_1(x, \hat{\beta})\right]}_0\right) \end{aligned}$$

This double robust estimator:

- Offers some protection against model misspecification
- will have smaller variance than the simple IPW estimator when  $P(x)$  holds.
- will have larger variance than the simple outcome if model is correct ( $M(x, z)$  is correct), but gives protection in the event it is not correct.

For variables that are unrelated to  $Z$  but related to the outcome

⇒ increase precision

⇒ should include them

For variables that are unrelated to the outcome but related to  $Z$

⇒ decrease precision

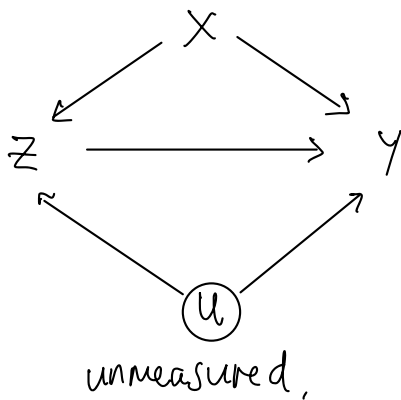
⇒ should not include them.



### 3.4: Unmeasured Confounding

So far, our models assumed no unmeasured confounding:

- Regression adjustment
- propensity score
- doubly robust,



E-value:

How **strong** would the unmeasured confounding have to be to negate the observed results?

→ quantify the minimum strength of association on the risk ratio scale that an unmeasured confounder must have with both the treatment + outcome to negate effect.

$E$  = evidence for causality.

## Features of E-value:

1. ranges from 1 to  $\infty$ .

Higher value  $\Rightarrow$  the stronger the unmeasured confounder must be to explain the observed association.

2. Different effect measures (RR, HR, risk difference, etc).

3. Ask this question of the 95% CI.

### Example:

D = binary outcome // 0 or 1

E = binary exposure // 0 or 1

C = measured confounders

U = unmeasured confounders.

$RR_{ED|C}^{obs}$  = observed / estimated relative risk of the exposure E on the outcome D, adjusted for measured confounders.

$$= \frac{P(D=1 | E=1, C=c)}{P(D=1 | E=0, C=c)} \quad \begin{array}{l} \text{// exposed, controlling for C} \\ \text{// not exposed, controlling for C} \end{array}$$

$RR_{ED}^{true}$  = true relative risk of exposure E on outcome D, adjusted for measured and unmeasured confounders.

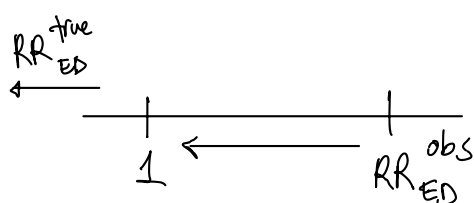
$$= \frac{\sum_{k=1}^K P(D=1 | E=1, C=c, U=k) P(U=k | C=c)}{\sum_{k=1}^K P(D=1 | E=0, C=c, U=k) P(U=k | C=c)}$$

$RR_{EU}$  = relative risk of the exposure E on the unmeasured confounder.

$RR_{UD}$  = RR of the unmeasured confounder on the outcome D

$$RR_{ED}^{obs} \neq RR_{ED}^{true}$$

$$RR_{ED}^{true} \geq RR_{ED}^{obs} / \underbrace{\frac{RR_{EU} RR_{UD}}{RR_{EU} + RR_{UD} - 1}}_{\text{joint bounding factor}}$$



$RR_{EU} + RR_{UD}$  = sensitivity parameters  $\rightarrow$  joint bounding factor.

$$RR_{EU} = RR_{UD}$$

E-value = minimum value for both associations that would be capable of attenuating the observed association to the null.

For risk ratio:

$$\text{E value} = RR_{ED}^{obs} + \sqrt{RR_{ED}^{obs} (RR_{ED}^{obs} - 1)}$$

// if  $RR_{ED}^{obs} < 1$ , then take the inverse before applying formula.

Formula can also be used for  
hazard ratio or odds ratio.